MiaPLC · example project

Belacqua

A sturgeon farm with 20 rearing tanks for 100 beluga sturgeons, controlled by MiaPLC: one PLC, two uLink I/O modules on the tank lines and a 10.1" touch panel. A real plant, not two lamps.

20
tanks, 100 fish, 1 t of biomass

4
MiaPLC devices on one Ethernet

369
ladder networks in 13 FCs

146 · 46
alarms · trend recordings

Manual version 1.0 · 05-11-2026 · projects in `MiaExamples/example_project/belacqua` · tested on real devices: **not yet**.

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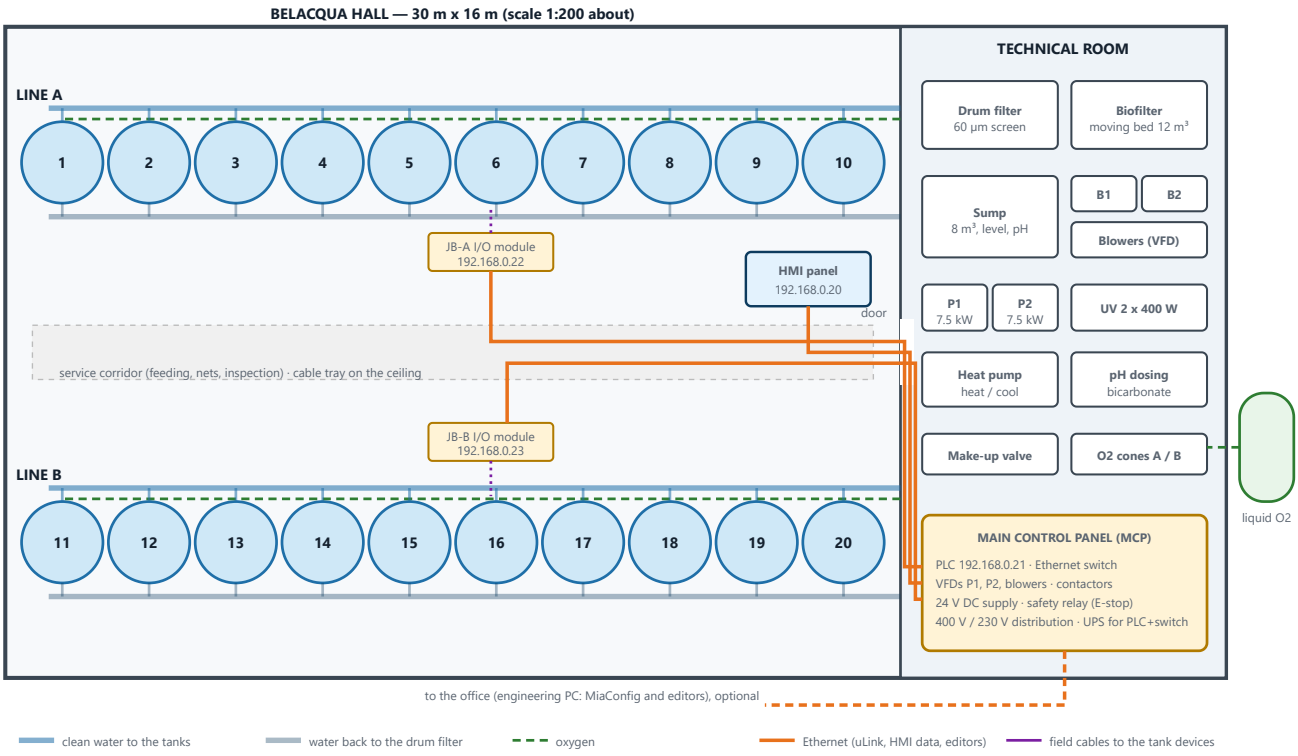
1. The plant
2. Hall layout
3. Network
4. Electrical design
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6. PLC program
7. HMI panel
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1. The plant

Beluga sturgeons (*Huso huso*) are bottom feeders that need cool, clean and well oxygenated water: 16–20 °C (best about 18 °C), dissolved oxygen above 7 mg/l, pH 7.0–8.0. Belacqua keeps 100 fish of about 10 kg, 5 per tank, in 20 round tanks of 3 m^3 (Ø 2 m, 1 m of water). All the tanks share one recirculating water loop (RAS): the water leaves the tanks, is cleaned and comes back; only a little new water is added every day.

Item	Size	Why
Tanks	20 x 3 m^3 = 60 m^3	1 t of fish at about 17 kg/m^3: comfortable for sturgeons
Circulation	60 m^3/h at night, 90 m^3/h by day, 120 m^3/h boost	1 to 2 water exchanges per hour in every tank
Pumps	2 x 7.5 kW with VFD, duty / standby	one pump is enough; the other covers faults and maintenance
Drum filter	120 m^3/h, 60 µm screen, automatic backwash	removes faeces and uneaten feed
Biofilter	moving bed 12 m^3, 2 blowers 2.2 kW with VFD	bacteria turn ammonia into nitrate
UV steriliser	2 x 400 W, 120 m^3/h	keeps bacteria and algae down
Heat pump	12 kW heating and cooling	holds 18 °C in winter and summer
Oxygen	liquid O2 tank, 2 oxygen cones, valve per tank	pure oxygen where and when a tank needs it
Feed	about 10 kg/day (1 % of the weight), 6 meals	portion computed per tank from fish and weight

2. Hall layout



Hall of 30 x 16 m: two lines of 10 tanks with the service corridor between them, the technical room with the water treatment and the main control panel. Each line has a junction box (JB-A, JB-B) in its middle with the uLink I/O module: the field cables of the 10 tanks stay short, only one Ethernet cable and the 24 V supply go back to the panel.

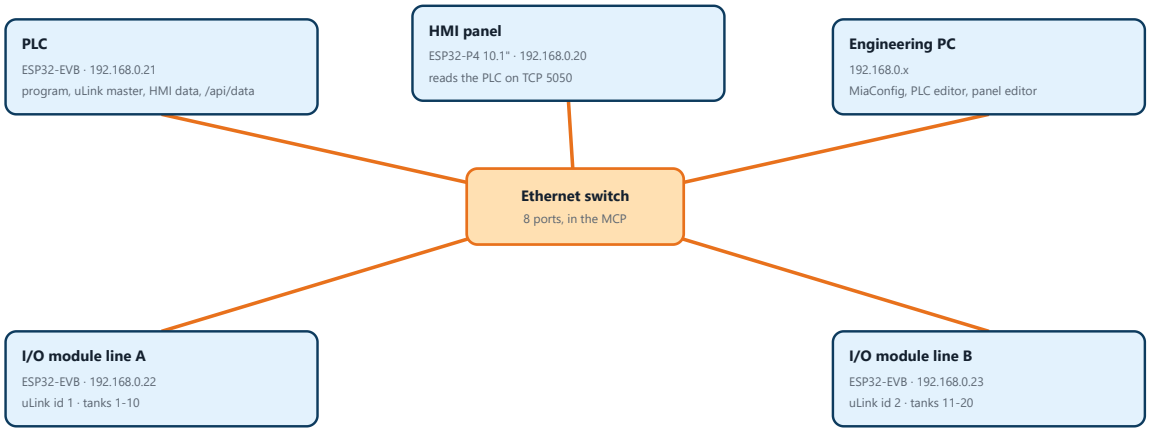
Where the devices are

- **Main control panel (MCP)**, technical room: PLC, Ethernet switch, VFDs, contactors, 24 V supply, safety relay, UPS.
- **JB-A, JB-B**: I/O module of the line, ADS1115 boards, input/output interface boards, terminals for the 10 tanks.
- **HMI panel**: at the end of the corridor, by the door of the technical room: the keeper sees all the tanks from there.

On every tank

- dissolved oxygen probe and temperature probe (4-20 mA);
- oxygen solenoid valve on the diffuser;
- automatic feeder with a hopper-empty sensor;
- overflow float and low-level float.

3. Network



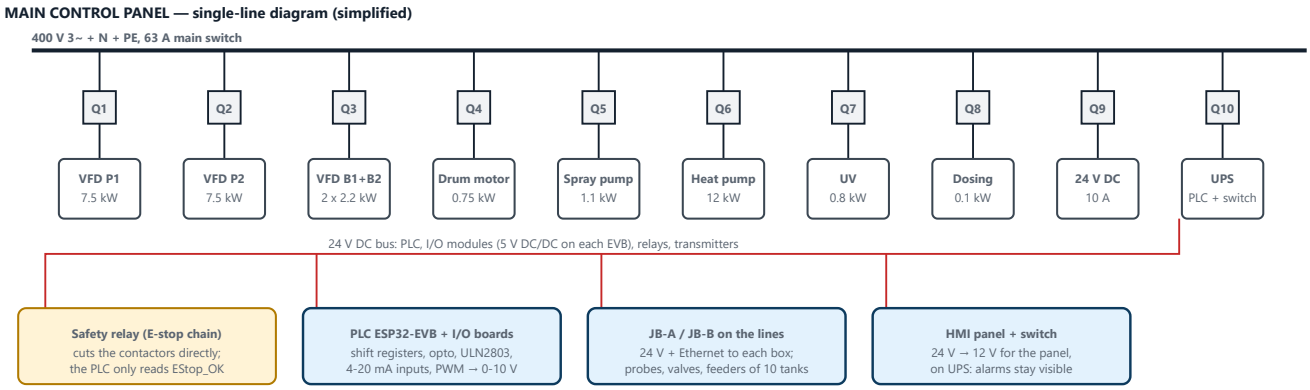
One cable per device. uLink exchanges the I/O of the modules every few milliseconds on the same network; the editors and the panel use TCP/IP beside it. Addresses: 192.168.0.0/24, set with MiaConfig (or by MAC).

All the devices on one Ethernet switch (192.168.0.0/24). uLink carries the I/O of the line modules; the panel reads the PLC on TCP 5050; MiaConfig and the editors reach every board, also by MAC when its address is unknown.

Device	Address	Project	Name on the network
PLC	192.168.0.21	`belacqua_plc_001`	`belacqua`
I/O module line A	192.168.0.22	`belacqua_io_002`	`linea` (uLink module and DB name)
I/O module line B	192.168.0.23	`belacqua_io_003`	`lineb`
HMI panel	192.168.0.20	`belacqua_hmi_010`	HMI-P4-xxxx

4. Electrical design

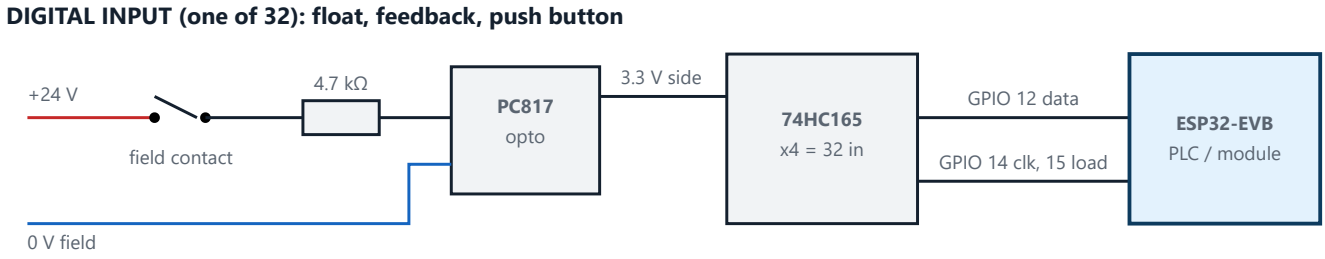
Safety. The emergency stop chain works through a safety relay that cuts the contactors and VFD enables directly. The PLC reads `EStop_OK` only to stop its logic and raise the alarm: safety never depends on the program. Follow the local electrical rules (EN 60204-1, IEC 61439) and have the panel checked.

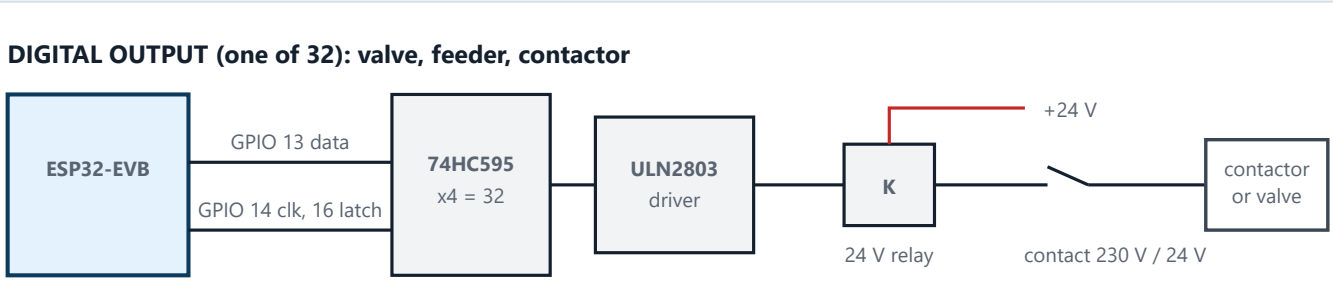


Main control panel. Motors on VFDs or contactors, everything controlled by the PLC through 24 V interface relays; PLC and switch on a small UPS, so alarms and data survive a power cut.

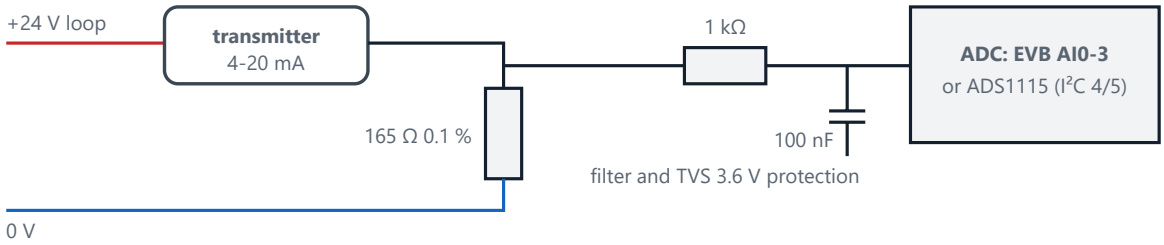
Connections of the controllers

The ESP32-EVB boards are used with their Ethernet only: the two relays of the board are not used. The digital I/O go through shift registers on small interface boards, the analog inputs through 4-20 mA shunts.





ANALOG INPUT: oxygen, temperature, pH, level, flow (4-20 mA, 2 wires)



660 mV = 4 mA = scale minimum · 3300 mV = 20 mA = scale maximum (set in the I/O table of the editor)

SPEED REFERENCE: PWM → 0-10 V → VFD of pumps and blowers



Board	Digital in	Digital out	Analog in	Analog out	Bus
PLC (MCP)	32 (4 x 74HC165)	32 (4 x 74HC595)	4 (EVB ADC)	4 PWM → 0-10 V	Ethernet LAN8720
I/O module line A, B (JB)	32	32	20 (4 EVB + 4 x ADS1115)	—	Ethernet, I²C SDA 4 / SCL 5

5. I/O lists

In the PLC program every point is a field of a DB generated from the I/O table: `HWDB` for the PLC board, `linea` and `lineb` for the modules. The analog values arrive already scaled (4-20 mA → engineering units) and averaged over 4 readings.

PLC (main control panel)

Type	Point	Program name	Device	Range
DI	`sin0.0`	`HWDB.EStop_OK`	emergency stop chain closed (NC)	
DI	`sin0.1`	`HWDB.Pump1_Running`	pump P1 feedback	
DI	`sin0.2`	`HWDB.Pump2_Running`	pump P2 feedback	
DI	`sin0.3`	`HWDB.Pump1_Fault`	pump P1 VFD fault	
DI	`sin0.4`	`HWDB.Pump2_Fault`	pump P2 VFD fault	
DI	`sin0.5`	`HWDB.Drum_HighWater`	drum filter high water float	
DI	`sin0.6`	`HWDB.Sump_LowLow`	sump low-low float	
DI	`sin0.7`	`HWDB.Blower1_Running`	blower B1 feedback	
DI	`sin0.8`	`HWDB.Blower2_Running`	blower B2 feedback	
DI	`sin0.9`	`HWDB.UV_Running`	UV steriliser lamps on	
DI	`sin0.10`	`HWDB.Chiller_Fault`	chiller fault contact	
DI	`sin0.11`	`HWDB.Heater_Fault`	heater fault contact	
DI	`sin0.12`	`HWDB.Dosing_Tank_Low`	bicarbonate tank low level	
DI	`sin0.13`	`HWDB.Ack_Button`	alarm acknowledge push button	
DI	`sin0.14`	`HWDB.Power_OK`	mains power monitor	
DI	`sin0.15`	`HWDB.Drum_Running`	drum filter motor feedback	
DO	`sout0.0`	`HWDB.Pump1_Run`	pump P1 VFD run	
DO	`sout0.1`	`HWDB.Pump2_Run`	pump P2 VFD run	
DO	`sout0.2`	`HWDB.Drum_Motor`	drum filter motor	
DO	`sout0.3`	`HWDB.Drum_Spray`	drum filter backwash spray pump	
DO	`sout0.4`	`HWDB.Blower1_Run`	biofilter blower B1	
DO	`sout0.5`	`HWDB.Blower2_Run`	biofilter blower B2	
DO	`sout0.6`	`HWDB.UV_On`	UV steriliser	
DO	`sout0.7`	`HWDB.Makeup_Valve`	make-up water valve	
DO	`sout0.8`	`HWDB.Heater_On`	heater (heat pump) enable	
DO	`sout0.9`	`HWDB.Chiller_On`	chiller enable	

Type	Point	Program name	Device	Range
DO	`sout0.10`	`HWDB.Dosing_Pump`	pH dosing pump	
DO	`sout0.11`	`HWDB.Horn`	alarm horn	
DO	`sout0.12`	`HWDB.Beacon`	alarm beacon	
AI	`vpin.34`	`HWDB.Sump_Level`	sump level, ultrasonic 4-20 mA	0…200 cm
AI	`vpin.35`	`HWDB.Water_Temp`	loop temperature after the heat exchanger, PT100 4-20 mA	0…40 C
AI	`vpin.36`	`HWDB.Water_pH`	pH probe in the sump, 4-20 mA	0…14 pH
AI	`vpin.39`	`HWDB.Flow_Rate`	circulation flow, magnetic flow meter 4-20 mA	0…150 m3/h
AO	`pwm.2`	`HWDB.Pump_Speed`	speed reference to the pump VFDs (0-100 %)	0…100 %
AO	`pwm.4`	`HWDB.Blower_Speed`	speed reference to the blower VFDs (0-100 %)	0…100 %
AO	`pwm.5`	`HWDB.Dosing_Speed`	stroke rate of the pH dosing pump (0-100 %)	0…100 %

Line modules (the same on line A and line B)

Program names: `T01_O2` , `T01_Temp` , `T01_FeedEmpty` , `T01_Overflow` , `T01_LowLevel` , `T01_O2Valve` , `T01_Feeder` … `T20_…` ; line signals `O2_Supply_OK` , `Aerator_Running` , `Light` , `Aerator` .

Tank	Oxygen (AI)	Temperature (AI)	Feeder empty · overflow · low level (DI)	O2 valve · feeder (DO)
tank 1 (A) / 11 (B)	`vpin.34`	`xadc1.2`	`sin0.0` · `sin0.1` · `sin0.2`	`sout0.0` · `sout0.1`
tank 2 (A) / 12 (B)	`vpin.35`	`xadc1.3`	`sin0.3` · `sin0.4` · `sin0.5`	`sout0.2` · `sout0.3`
tank 3 (A) / 13 (B)	`vpin.36`	`xadc2.0`	`sin0.6` · `sin0.7` · `sin0.8`	`sout0.4` · `sout0.5`
tank 4 (A) / 14 (B)	`vpin.39`	`xadc2.1`	`sin0.9` · `sin0.10` · `sin0.11`	`sout0.6` · `sout0.7`
tank 5 (A) / 15 (B)	`xadc0.0`	`xadc2.2`	`sin0.12` · `sin0.13` · `sin0.14`	`sout0.8` · `sout0.9`
tank 6 (A) / 16 (B)	`xadc0.1`	`xadc2.3`	`sin0.15` · `sin0.16` · `sin0.17`	`sout0.10` · `sout0.11`
tank 7 (A) / 17 (B)	`xadc0.2`	`xadc3.0`	`sin0.18` · `sin0.19` · `sin0.20`	`sout0.12` · `sout0.13`
tank 8 (A) / 18 (B)	`xadc0.3`	`xadc3.1`	`sin0.21` · `sin0.22` · `sin0.23`	`sout0.14` · `sout0.15`
tank 9 (A) / 19 (B)	`xadc1.0`	`xadc3.2`	`sin0.24` · `sin0.25` · `sin0.26`	`sout0.16` · `sout0.17`
tank 10 (A) / 20 (B)	`xadc1.1`	`xadc3.3`	`sin0.27` · `sin0.28` · `sin0.29`	`sout0.18` · `sout0.19`
line	—	—	`sin0.30` O2 supply OK · `sin0.31` aerator running	`sout0.20` lights · `sout0.21` emergency aerator

6. PLC program

Ladder (LD) in 13 functions called by Main, 369 networks, about 30 KB of program, at most 3300 instructions per scan. Data in 6 DBs: `Settings` and `TankSettings` (retentive: they survive a power cut), `Process`, `Tanks`, `Alarms`, `Stats` (retentive).

FC	Function	What it does
1	Clock and schedules	Local time from the PLC clock, day of the week, day / night, peak tariff, counters of the day.
2	Water level	Make-up valve with hysteresis; low-low level stops pumps and heating.
3	Recirculation	Flow target by day, at night (lower) and with a water alert (higher); the pump speed follows the flow in 5 s steps, so the pumps run at the lowest speed that gives it. Duty/standby pumps swap every 24 h and on a fault.
4	Drum filter	Backwash on high water, every N minutes and every night at a set time, when energy costs less.
5	Biofilter and UV	Blower 1 always, blower 2 by day or with a water alert, blower speed day / night; UV only with flow.
6	Oxygen	Per tank: oxygen valve with hysteresis around the set point; low and critical oxygen per line; emergency aerator of the line while a tank is critical.
7	pH dosing	Bicarbonate in pulses (on / pause) only with flow; timeout alarm if the pH does not rise.
8	Temperature	Heat pump in heating or cooling with a band; wider band in the peak tariff hours; never both together.
9	Feeding	Up to 8 meals a day, a fasting day, portion per tank = fish x weight x daily % / meals; a tank with low oxygen or bad temperature skips the meal; tank lights per line on the chosen days.
10	Tank alarms	An alarm word per tank: O2 low, O2 critical, feeder empty, overflow, low level, temperature.
11	Plant alarms	26 latched alarms; acknowledge from the panel or the push button; horn until acknowledged, beacon.
12	Tank settings	The panel edits one tank at a time (in service, modes, fish, weight) and writes it back.
13	Statistics	Running hours, backwashes, meals, feed given and estimated pump energy of the day.

> Every actuator has a mode: **Auto** (the program decides), **Manual ON**, **Manual OFF**. Interlocks stay active in manual: no pump without level, no UV or heating without flow, no feeder with an empty hopper.

Energy saving, in numbers

A centrifugal pump uses power with the cube of its speed: at 70 % speed it takes about 34 % of the power. Running the pumps at the lowest speed that gives the flow, and at 60 m^3/h instead of 90 m^3/h for the 11 hours of night, saves roughly 40 % of the pumping energy of the day. The blowers slow down at night and the second one stops; heating and cooling wait for a wider band in the peak tariff hours; the drum filter washes at 03:00.

Data for other systems

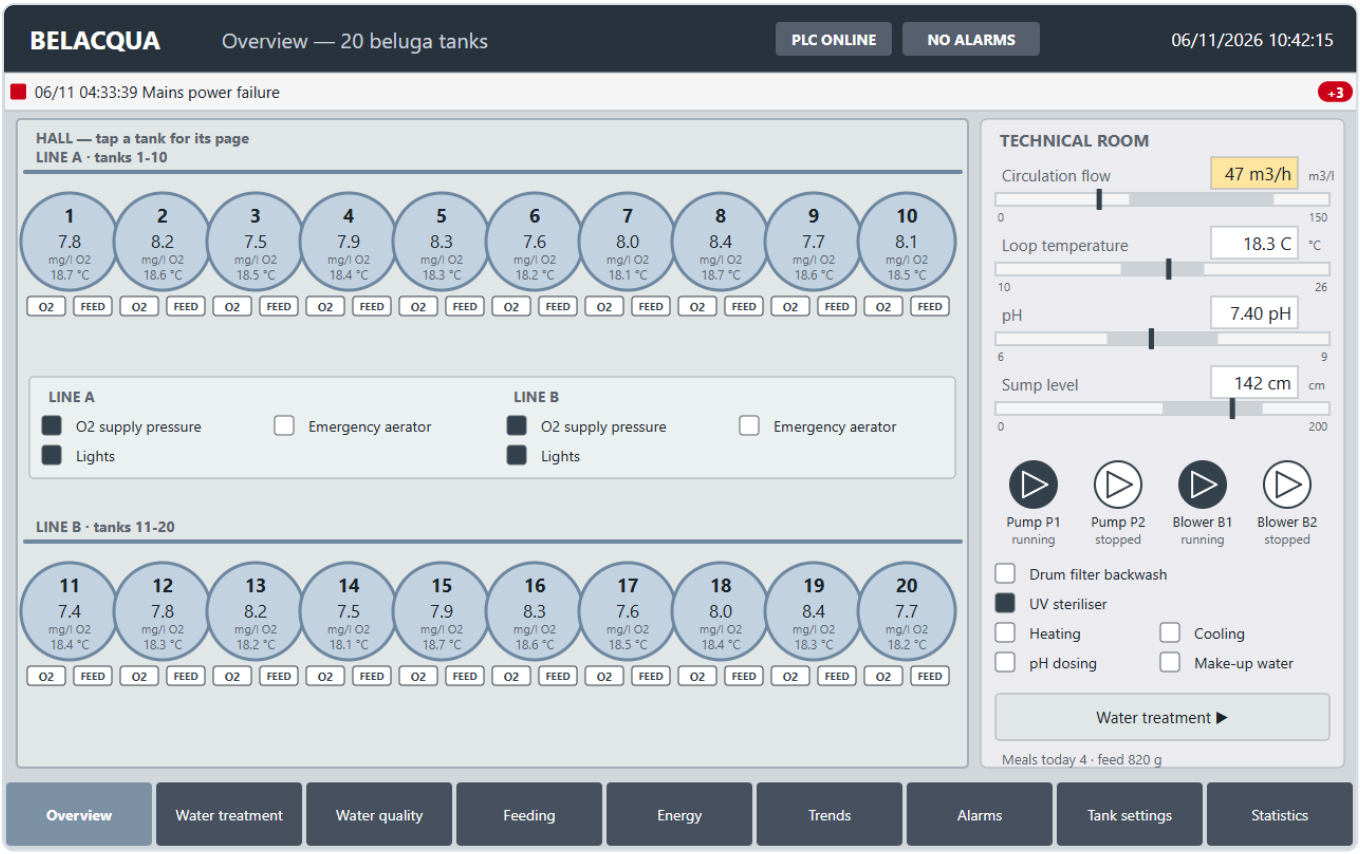
With **data exposure** the PLC publishes oxygen, temperature and alarms of every tank and the plant values as JSON (`http://192.168.0.21/api/data`), and accepts four set points: a farm management software can read the plant without touching the program.

7. HMI panel

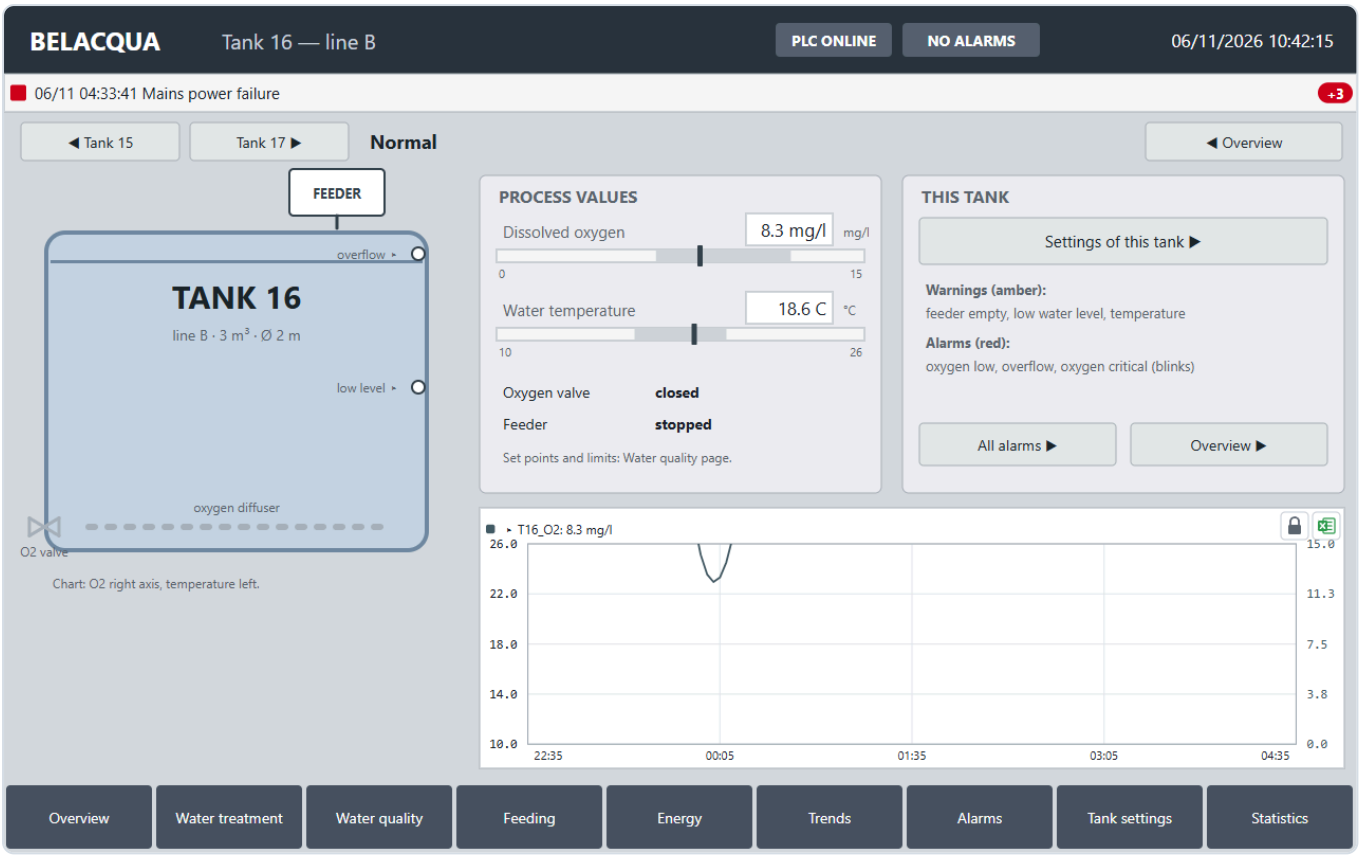
The panel follows the high-performance HMI rules of ISA-101: neutral grey screens, colour only when something is wrong (amber warning, red alarm, blinking for a critical alarm), analog values as a pointer on a scale with the normal band in darker grey, equipment drawn filled when it runs and hollow when it is stopped, and the same header, alarm line and navigation on every page. Three levels of pages:

Level	Page	Content
1	Overview	the hall: the 20 tanks where they are, each with oxygen, temperature, O2 and FEED marks, amber or red when the tank has a warning or an alarm; the technical room with flow, temperature, pH, level, pumps and blowers
2	Water treatment	the recirculation loop drawn as a flow diagram, values, modes of pumps, drum filter, blowers, UV, make-up
2	Water quality	oxygen of all 20 tanks as bars on the normal band, oxygen, pH and temperature set points and modes
2	Feeding · Energy · Trends · Alarms · Tank settings · Statistics	meal times, schedules and energy saving, 6-hour trends, alarm list and history, settings of one tank at a time, running hours and totals of the day
3	Tank 1 … Tank 20	one page per tank: the tank drawn with feeder, oxygen diffuser and valve, the most severe alarm in words, oxygen and temperature indicators, the 6-hour trend of the tank, previous / next tank

The screens



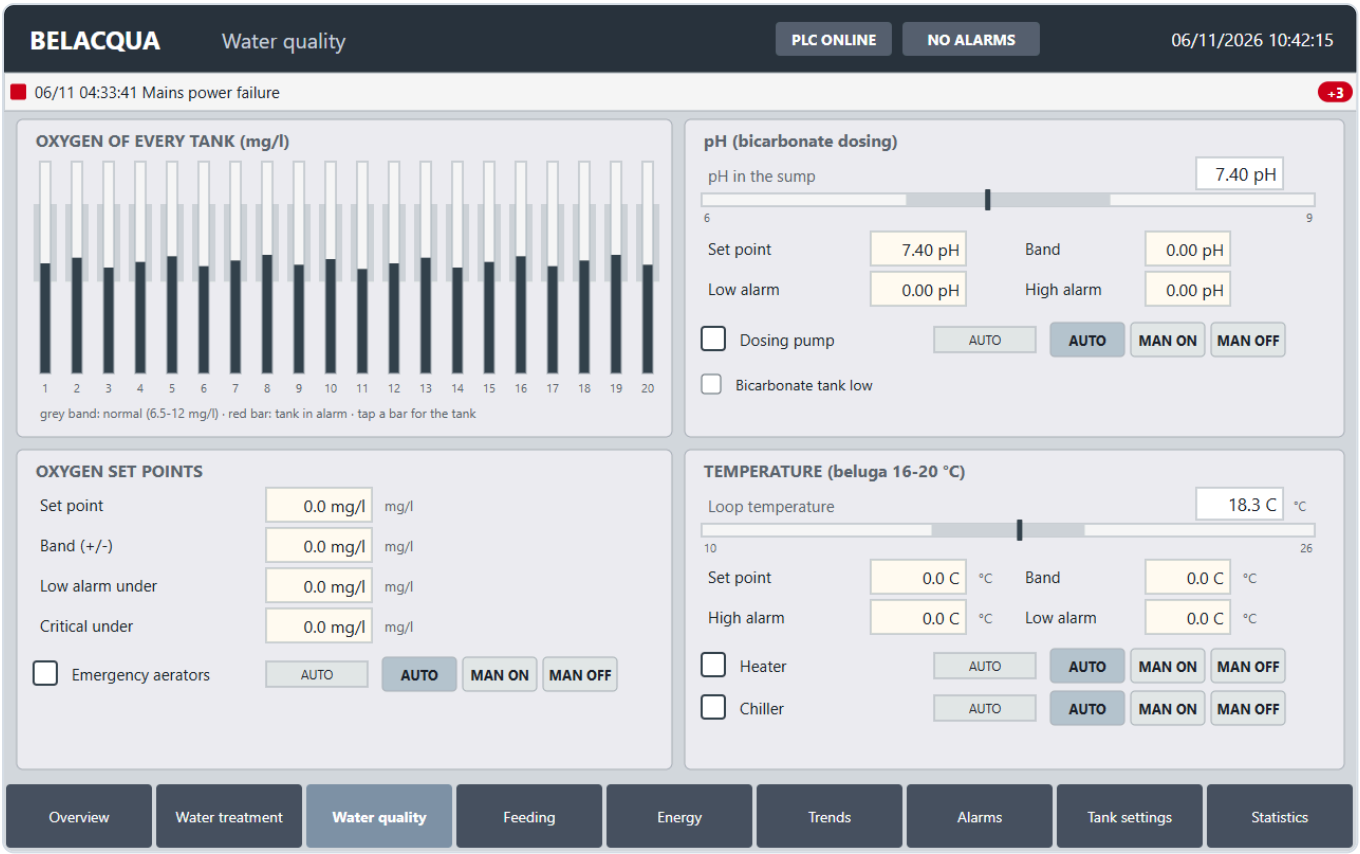
Level 1, overview: the 20 tanks where they are in the hall. Grey-blue is normal; tank 4 and 12 have a warning (amber), tank 9 an alarm (red), tank 16 an oxygen critical alarm (red, blinking: here caught while off). O2 and FEED under a tank: its oxygen valve or feeder is on. Right: the technical room with flow, loop temperature, pH and level on scales with the normal band, pumps and blowers filled when running.



Level 3, the page of a tank: the most severe alarm in words, the tank drawn with feeder, oxygen diffuser and valve, oxygen and temperature on their scales (the value turns amber outside the normal band), the 6-hour trend of the tank (empty in this picture: no data yet).



Level 2, water treatment: the recirculation loop, values and the modes of the machines (AUTO, MAN ON, MAN OFF; amber when not in automatic).



Level 2, water quality: the oxygen of all 20 tanks on the normal band, set points of oxygen, pH and temperature.

Pictures made with sample values (no field devices): on the plant the values are the real ones.

The panel talks to the PLC through the **HMI data** table of the PLC program (240 of 256 rows: 4 per tank plus the plant). It needs the panel firmware 0.1.8 or later (64 recordings, 20 series per chart).

8. Alarms

26 plant alarms, latched in the PLC until acknowledged; plus 6 alarms per tank (feeder empty, low water level, temperature out of range, O2 low, overflow, O2 critical) from the alarm word of the tank: 146 alarms on the panel. Red: alarm (act now); orange: warning.

No.	Plant alarm	PLC field
1	Emergency stop pressed	`Al_EStop`
2	Mains power failure	`Al_Power`
3	Pump P1 fault	`Al_Pump1`
4	Pump P2 fault	`Al_Pump2`
5	No circulation flow	`Al_No_Flow`
6	Sump level low-low: pumps stopped	`Al_Level_LowLow`
7	Drum filter screen clogged	`Al_Drum_Clogged`
8	Drum filter motor fault	`Al_Drum_Motor`
9	Biofilter blower fault	`Al_Blowers`
10	UV steriliser fault	`Al_UV_Fault`
11	UV lamps to replace	`Al_UV_Service`
12	Oxygen supply pressure low, line A	`Al_O2_Supply_A`
13	Oxygen supply pressure low, line B	`Al_O2_Supply_B`
14	Emergency aerator line A not running	`Al_Aerator_A`
15	Emergency aerator line B not running	`Al_Aerator_B`
16	pH too low	`Al_pH_Low`
17	pH too high	`Al_pH_High`
18	pH dosing too long: check the bicarbonate tank	`Al_pH_Dose_Timeout`
19	Bicarbonate tank low	`Al_Dosing_Tank_Low`
20	Water temperature high	`Al_Temp_High`
21	Water temperature low	`Al_Temp_Low`
22	Chiller fault	`Al_Chiller`
23	Heater fault	`Al_Heater`
24	I/O module line A (uLink) not answering	`Al_Module_A`
25	I/O module line B (uLink) not answering	`Al_Module_B`
26	PLC clock not set: feeding and schedules stopped	`Al_Clock`

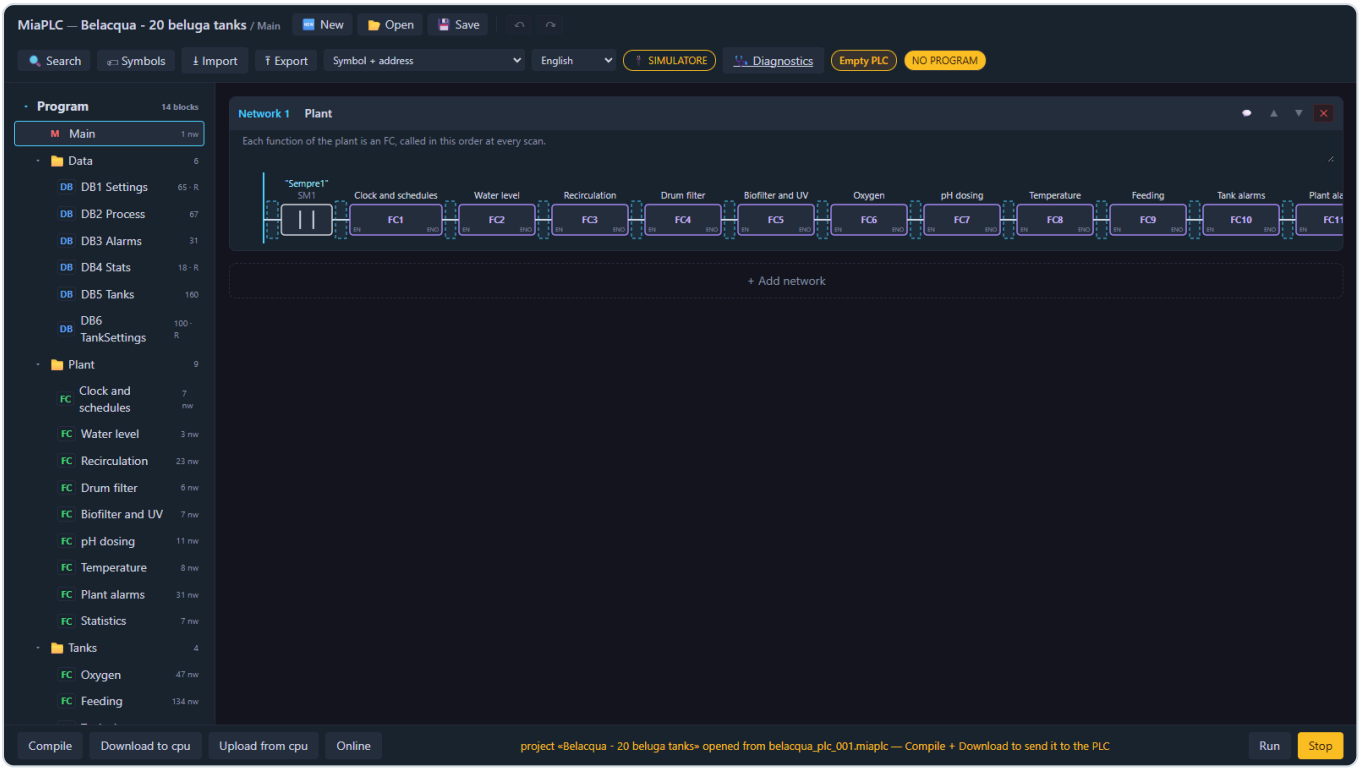
9. Settings

All in the retentive DB `Settings`, editable on the panel within limits.

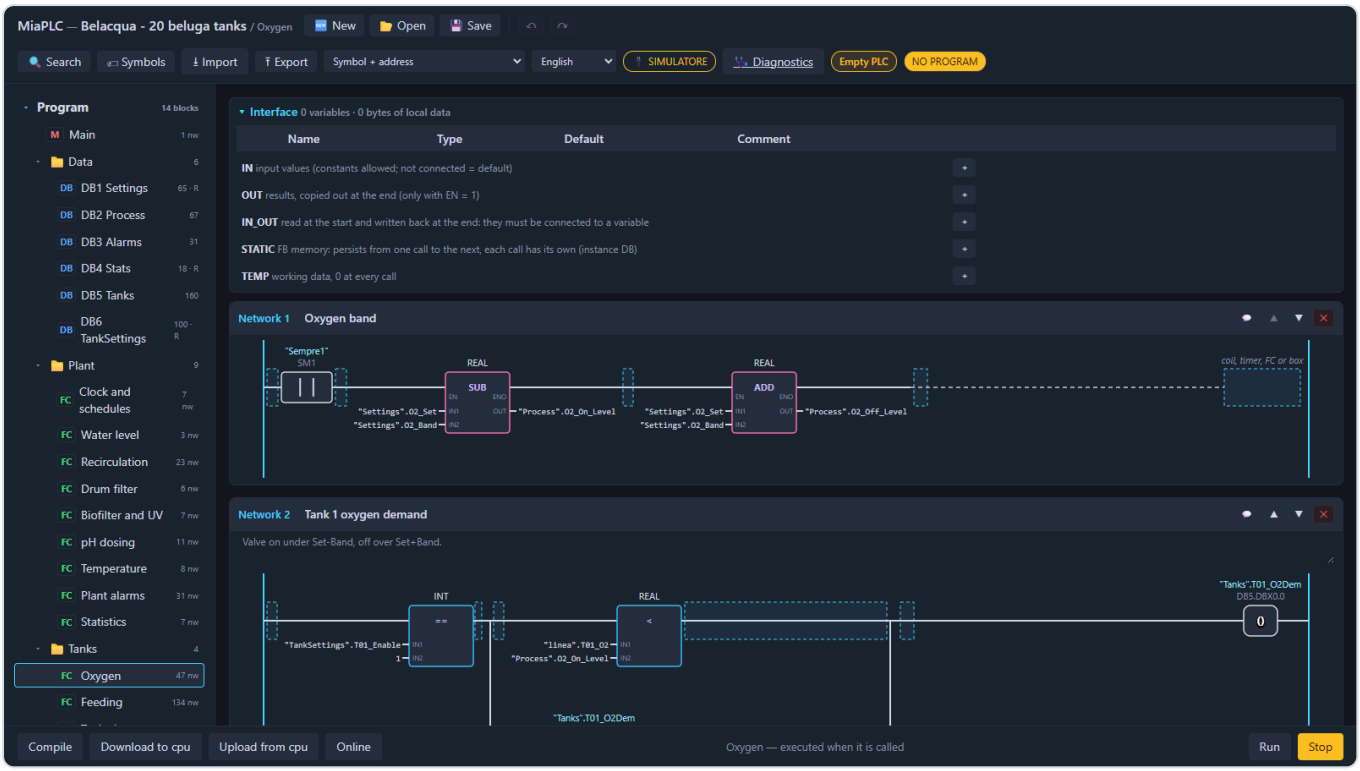
Name	Type	Default	Meaning
`Level_Makeup_On`	REAL	120.0	sump level (cm) under which the make-up valve opens
`Level_Makeup_Off`	REAL	130.0	sump level (cm) at which the make-up valve closes
`Level_LowLow`	REAL	80.0	sump level (cm) under which pumps and heating stop
`Flow_Day`	REAL	90.0	circulation flow by day (m3/h): 20 tanks x 3 m3 x 1.5 exchanges per hour
`Flow_Night`	REAL	60.0	circulation flow at night (m3/h): energy saving, fish at rest
`Flow_Boost`	REAL	120.0	circulation flow when the water is not good (low oxygen, temperature)
`Flow_Band`	REAL	3.0	flow band (m3/h): the speed changes only outside it
`Speed_Step`	REAL	1.0	pump speed change every 5 s (%)
`Speed_Min`	REAL	35.0	minimum pump speed (%)
`Speed_Max`	REAL	100.0	maximum pump speed (%)
`Flow_Min`	REAL	30.0	flow (m3/h) under which there is no circulation
`Pump_Swap_s`	UDINT	86400	running time (s) after which the duty pump changes
`Blower_Day`	REAL	85.0	blower speed by day (%)
`Blower_Night`	REAL	60.0	blower speed at night (%): one blower only
`Drum_Interval`	TIME	1800000	drum filter backwash every ... (ms) when the float does not ask before
`Drum_Wash`	TIME	30000	duration of one backwash (ms)
`Drum_Max_Hour`	INT	8	backwashes in one hour above which the screen is clogged
`Drum_Night_Wash`	UDINT	180	scheduled backwash at ... (minute of the day, 180 = 03:00, cheap energy)
`O2_Set`	REAL	8.0	dissolved oxygen set point (mg/l)
`O2_Band`	REAL	0.5	oxygen valve band (mg/l): on under Set-Band, off over Set+Band
`O2_Low`	REAL	6.5	low oxygen in a tank (mg/l): alarm, no feeding in that tank, flow boost
`O2_Critical`	REAL	5.0	critical oxygen (mg/l): emergency aerator of the line on
`pH_Set`	REAL	7.4	pH set point
`pH_Band`	REAL	0.1	pH dosing band: dosing under Set-Band until Set+Band
`pH_Low`	REAL	6.8	low pH alarm
`pH_High`	REAL	8.2	high pH alarm
`pH_Dose_On`	TIME	2000	dosing pulse (ms)

Name	Type	Default	Meaning
`pH_Dose_Pause`	TIME	20000	pause between two pulses (ms): mixing time
`pH_Dose_Max`	TIME	900000	pH still low after dosing for ... (ms): alarm
`Dosing_Speed_Set`	REAL	50.0	stroke rate of the dosing pump (%)
`Temp_Set`	REAL	18.0	water temperature set point (C), beluga: 16-20 C
`Temp_Band`	REAL	0.5	temperature band off-peak (C)
`Temp_Band_Peak`	REAL	1.5	temperature band in the peak tariff hours (C): energy saving
`Temp_High`	REAL	21.0	high temperature alarm (C)
`Temp_Low`	REAL	14.0	low temperature alarm (C)
`Peak_Start`	UDINT	480	peak tariff from ... (minute of the day, Monday-Friday)
`Peak_End`	UDINT	1140	peak tariff until ... (minute of the day)
`Day_Start`	UDINT	420	day from ... (minute of the day): lights, day flow
`Day_End`	UDINT	1200	night from ... (minute of the day)
`Light_Days`	INT	127	lights on these days: bit 0 Monday ... bit 6 Sunday (127 = every day)
`Feed_Enable`	BOOL	True	automatic feeding enabled
`Fast_Day`	INT	6	fasting day, no feeding: 0 Monday ... 6 Sunday, 7 = none
`Feed_Pct`	REAL	1.0	daily feed in % of the fish weight
`Feeder_Rate`	REAL	5.0	feeder output (g/s), the same model on every tank
`Meal_1`	UDINT	480	meal 1: minute of the day (480 = 08:00), 9999 = off
`Meal_2`	UDINT	600	meal 2: minute of the day (480 = 08:00), 9999 = off
`Meal_3`	UDINT	720	meal 3: minute of the day (480 = 08:00), 9999 = off
`Meal_4`	UDINT	840	meal 4: minute of the day (480 = 08:00), 9999 = off
`Meal_5`	UDINT	960	meal 5: minute of the day (480 = 08:00), 9999 = off
`Meal_6`	UDINT	1080	meal 6: minute of the day (480 = 08:00), 9999 = off
`Meal_7`	UDINT	9999	meal 7: minute of the day (480 = 08:00), 9999 = off
`Meal_8`	UDINT	9999	meal 8: minute of the day (480 = 08:00), 9999 = off
`UV_Service_h`	REAL	9000.0	UV lamp hours before replacement
`Time_Offset_s`	UDINT	3600	local time offset from UTC (s): 3600 winter, 7200 summer (Italy)

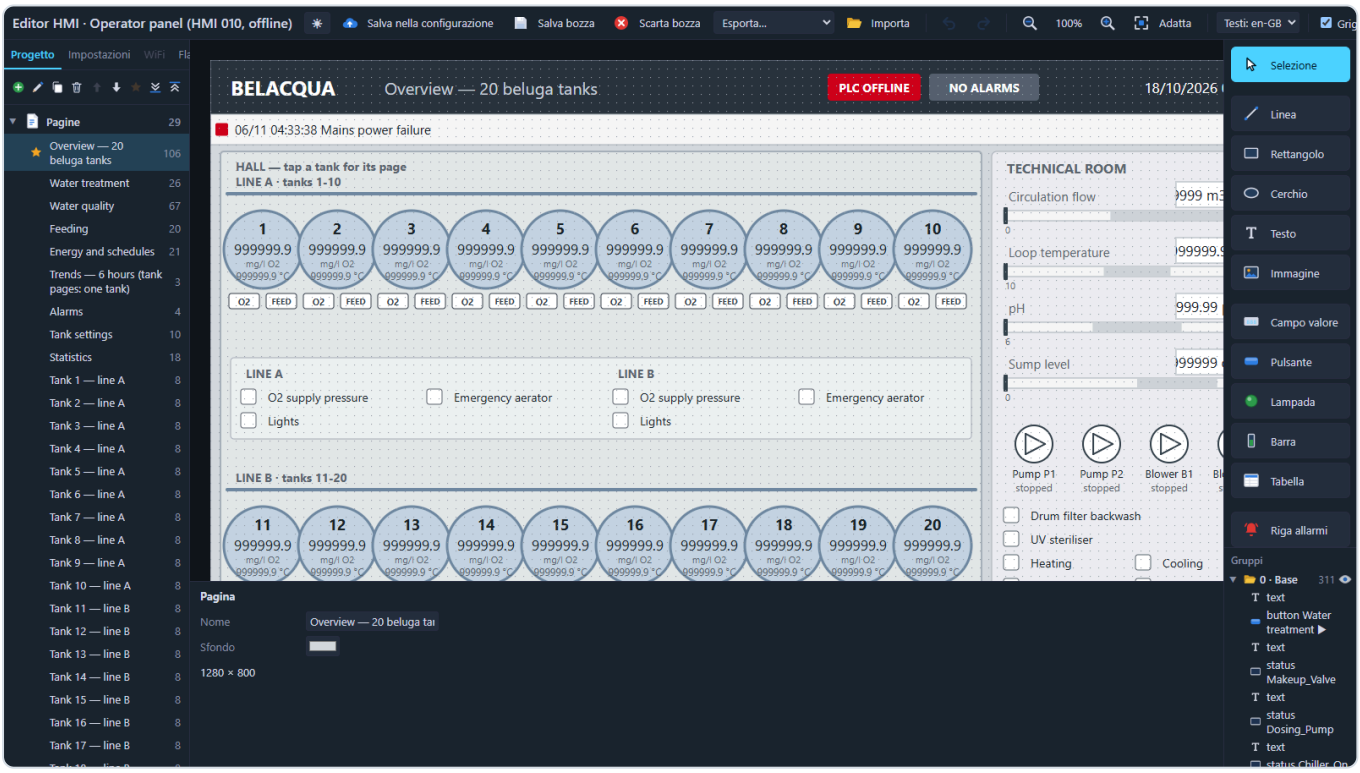
10. Build and commissioning



The MiaPLC editor (in the browser, served by the PLC itself) with the Belacqua program: on the left the blocks in folders (data, plant, tanks, system DBs of the board and of the two uLink modules), in the middle Main calling the 13 functions.



The ladder of FC6 Oxygen: the band around the set point, then for each tank the oxygen demand with hysteresis and the valve.



The panel editor (in the browser, served by the panel): the 29 pages on the left, the page in the middle, the objects on the right; Importa loads `belacqua_hmi_010.miahmi`, Pubblica sul pannello makes it the project of the panel.

1. **Firmware**: open each `.mia` in MiaConfig, *Genera → Compila firmware*, load it on the board by USB (first time) or over the network. The PLC project uses developer mode (needed for the HMI data).

2. **Network**: every board starts with the address of its project; a board that is not reachable can be found and set by MAC in MiaConfig (*Schede in rete*).
3. **Modules**: on the PLC, Diagnostics → I/O modules: pair `linea` and `lineb`.
4. **PLC program**: open `belacqua_plc_001.miaplc` in the PLC editor, connect to 192.168.0.21, check the I/O table against the real board, Compile, Download with the PLC in STOP, then RUN.
5. **Panel**: in the panel editor (http://192.168.0.20) Importa `belacqua_hmi_010.miahmi`, publish; in the panel settings set the PLC address 192.168.0.21.
6. **Clock**: set the PLC date and time from the panel (Energy page): schedules and feeding wait for it.
7. **Checks**: each tank in Manual ON / OFF for valve and feeder, probes against a hand meter, every alarm once, emergency stop, power cut with the UPS.

11. Test status

Part	Checked	On real devices
MiaConfig projects (4)	MiaConfig checks: no errors	no
PLC program	editor compiler: no errors, no warnings	no
HMI panel project	panel project checks (firmware 0.1.8): valid	no
uLink module numbers	the program reads `ulink_ok` bit 1 (line A) and bit 2 (line B)	to check
Electrical drawings	example design, to adapt to the real plant and its rules	—